Mathematical Statistics
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# Causal Modeling with Stationary Diffusions

## Efficient Learning with Stein-type Discrepancies

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## Stationary Diffusions

Causal models aim to represent observational and interventional distributions.
Stationary diffusions model a $d$-dimensional random variable $X$ as the **stationary distribution** $P$ of an SDE

$$dX_t = b(X_t)\,dt + \sigma(X_t)\,dB_t, \tag{1}$$

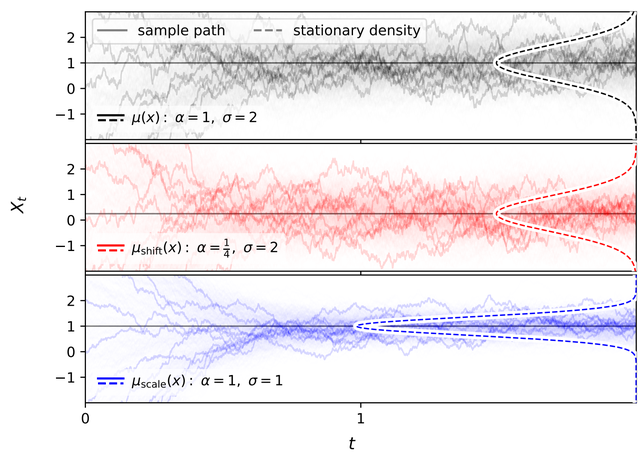
with drift $b$ and diffusion $\sigma$. **Interventions** alter the SDE. We consider shift-scale interventions, i.e.,

$$\tilde{b}(x)_i = b(x)_i + \delta_i \text{ and } \tilde{\sigma}(x)_i = \beta_i \sigma(x)_i. \tag{2}$$

**Goal.** Learn drift, diffusion, and interventional parameters to represent observational and interventional distributions, and generalize to unseen interventions.

**Example (Ornstein–Uhlenbeck).**

$$dX_t = b(X_t - \alpha)\,dt + \sigma\,dB_t \quad \Rightarrow \quad X \sim \mathcal{N}(\alpha, -\tfrac{\sigma^2}{2b})$$




**Key fact.** For a target distribution $P$, one can construct SDEs whose stationary distribution equals $P$.

**This work.** Given $b$, $\sigma$, we propose a **deviation measure** between $P$ and the induced stationary distribution, inspired by *KDS* (Lorch et al., 2024).

**Idea.** Build on Stein discrepancies. These use operators $\mathcal{T}_P$ such that

$$\sup_{g \in \mathcal{G}} \left| \mathbb{E}_{X \sim Q}[(\mathcal{T}_P g)(X)] \right| = 0 \quad \Longleftrightarrow \quad P = Q. \tag{3}$$

## Why Stationary Diffusions?

- Natural model class for interacting dynamical systems.
- Interventions correspond to *mechanism changes*, not merely conditioning.
- Handles feedback loops, by unfolding in time; not requiring acyclic structure.
- Data is collected from equilibrated systems; data is independent of time.

**Limits:**

- Counterfactuals are not naturally represented.
- Mechanisms are less interpretable than SCMs.

## Characterization of Stationarity

1. **Stationarity.** A probability distribution $P$ is a stationary solution of the SDE (1) if and only if

$$\left.\frac{d}{dt}\right|_{t=0} \mathbb{E}[f(X_t)] = \mathbb{E}_{X \sim P}[\mathcal{L}f(X)] = 0 \quad \text{for all } f \in C^2(\mathbb{R}^d), \tag{4}$$

meaning, the process $f(X_t)$ has zero average instantaneous change when $X_t \sim P$.

**Definition 1** (Generator)**.** For a test function $f \in C^2(\mathbb{R}^d)$, the **infinitesimal expected change** of the process $f(X_t)$ starting in $x$, is given by

$$\mathcal{L}f(x) := \lim_{t \downarrow 0} \frac{\mathbb{E}[f(X_t) \mid X_0 = x] - f(x)}{t} = \langle b(x), \nabla f(x) \rangle + \frac{1}{2} \operatorname{tr}\left(\sigma(x)\sigma(x)^T \nabla^2 f(x)\right). \tag{5}$$

2. **Kernel Deviation from Stationarity (Lorch et al. 2024).**

**(1) Stationarity condition** $\longrightarrow$ **(2) RKHS reformulation** $\longrightarrow$ **(3) Closed-form Maximizer in** $\mathcal{H}$

$$\sup_{f \in C^2(\mathbb{R}^d)} \mathbb{E}_{X \sim P}[\mathcal{L}f(X)] = 0 \qquad \sup_{h \in \mathcal{H}} \mathbb{E}_{X \sim P}[\mathcal{L}h(X)] = 0 \qquad \mathbb{E}_{X \sim P, Y \sim P}\left[\mathcal{L}_1 \mathcal{L}_2 k(X, Y)\right] = 0$$

## Stein-type Kernel Deviation from Stationarity

1. **SKDS Operator.** Define the operator $\mathcal{S}$ on $g \in C^1(\mathbb{R}^d, \mathbb{R}^d)$ by

$$\mathcal{S}(g)(x) := \langle b(x), g(x) \rangle + \frac{1}{2} \operatorname{tr}\left(\sigma(x)\sigma(x)^\top \nabla_x g(x)\right). \tag{6}$$

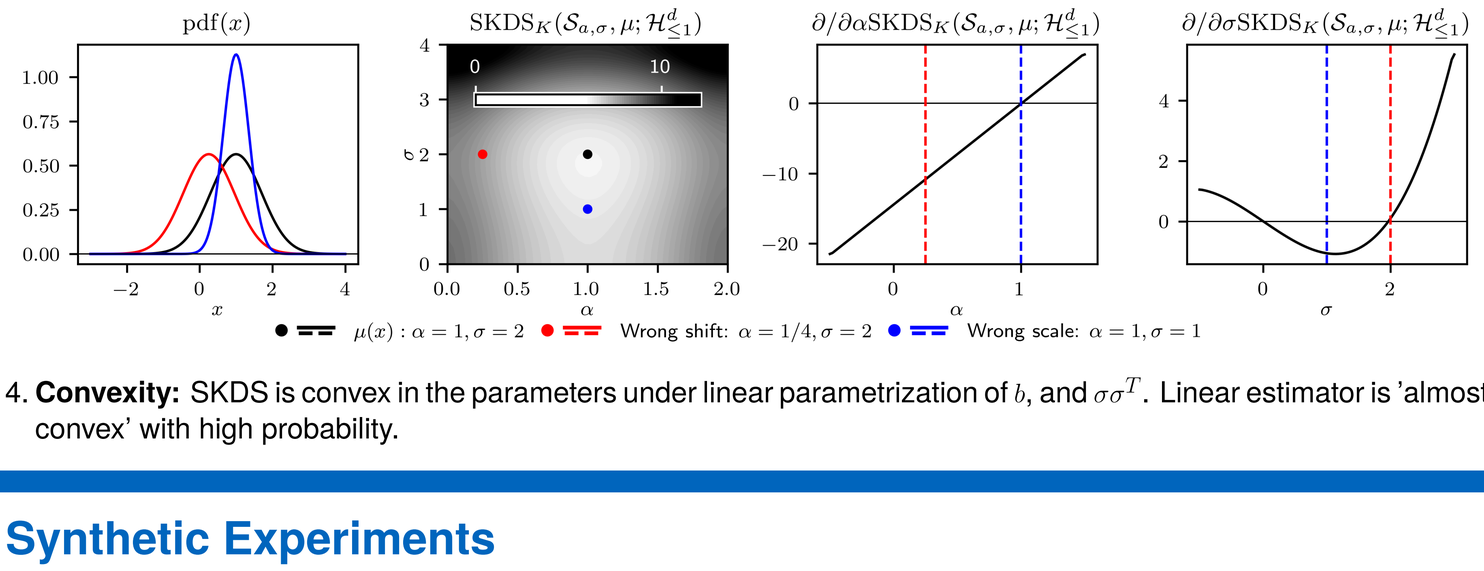
In (4): Replace $\mathcal{L}$ with $\mathcal{S}$ and maximize over RKHS of vector-valued functions $g \in \mathcal{H}^d$ to receive the *Stein-type Kernel Deviation from Stationarity with closed-form*:

$$\mathrm{SKDS}(\mathcal{S}, P; \mathcal{H}^d_{\leq 1}) = \mathbb{E}_{X \sim P} \mathbb{E}_{Y \sim P}\left[\mathcal{S}_1 \mathcal{S}_2 K(X, Y)\right]. \tag{7}$$

2. **Characterization.** A diffusion (1) is stationary and reversible with marginal $P$ iff

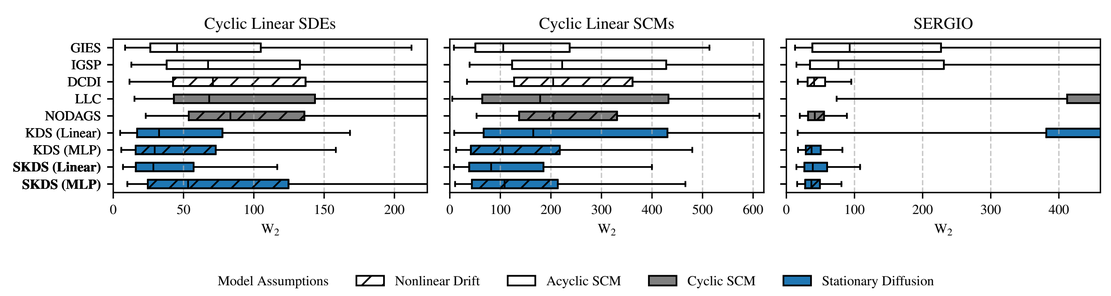
$$\mathrm{SKDS}(\mathcal{S}, P; \mathcal{H}^d_{\leq 1}) = 0. \tag{8}$$

3. **Learning.** Minimize SKDS over parametrized SDEs to fit observational and interventional distributions.




4. **Convexity:** SKDS is convex in the parameters under linear parametrization of $b$, and $\sigma\sigma^T$. Linear estimator is 'almost convex' with high probability.

## Synthetic Experiments




**Benchmark.** Erdős–Rényi causal graphs with $d = 20$ variables. Performance measured by Wasserstein-2 distance under $10$ unseen interventions across $50$ random systems.

**Results.** Empirically, SKDS **matches or improves** upon KDS and other competitive baselines across linear and nonlinear, cyclic and acyclic settings, while achieving a **5x reduction in computation time** against KDS.

**Future Work**

- Strengthen the link from SKDS-based SDE estimation to causal discovery.
- Many causal questions remain open for stationary diffusions.

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**Acknowledgement**